# OmniMed-FL: A Robust Multimodal Federated Learning Framework for Clinical Diagnosis

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***Abstract*—We present OmniMed-FL, a controlled systems study of multimodal federated learning for five-class image–text classification. The proxy corpus contains 2,400 public chest radiographs, 600 synthetic COVID-19 image surrogates, and 3,000 class-conditioned synthetic notes paired to images by class rather than patient. Public DistilBERT and Vision Transformer Base/16 checkpoints initialize the encoders, while task modules are initialized randomly; no operational run starts from a model trained on pooled client data. We vary Dirichlet skew, client count, anti-collapse components, initialization, eight operational fusion rules, missing text, and matched federated controls, while reporting training time, peak allocated graphics processing unit (GPU) memory, and formula-derived 32-bit floating-point (FP32) bidirectional communication volume. At $K{=}5$ and severe skew ($\alpha{=}0.1$), local-only training obtains 0.310 Macro F1 (five-client mean), versus $0.652{\pm}0.057$ for FedAvg, $0.685{\pm}0.064$ for FedProx, and $0.192{\pm}0.129$ for our SCAFFOLD-style AdamW adaptation; the 0.033 FedProx–FedAvg gap is smaller than either two-seed sample standard deviation (SD). In the heterogeneity sweep, FedAvg changes from $0.896{\pm}0.004$ at $\alpha{=}5$ to $0.674{\pm}0.185$ at $\alpha{=}0.1$. A branch audit records 0.248/0.324/0.573 GiB one-way FP32 model-state sizes for text/image/multimodal models. Because all notes and the 600 COVID-19 images are synthetic, image–text pairing is not patient-level, and no external clinical validation is performed, these are descriptive proxy comparisons rather than estimates of diagnostic performance, privacy, or deployment readiness.**

***Index Terms*—Multimodal Federated Learning, Non-Identically Distributed Data, Multimodal Fusion, Clinical Classification, Communication Cost, Retrieval**

## I. Introduction

Clinical prediction often draws on images and textual context. Centralized image classifiers [1], [2] analyze radiographs, whereas medical vision–language models can combine images with text. Centralized cross-institutional training, however, ordinarily requires data pooling. Federated learning (FL) instead exchanges parameter updates while retaining each client's raw examples [3]. This data-locality property is useful, although FL alone is not a privacy guarantee and does not remove risks from model updates.

Federated medical imaging [4]–[6] and centralized medical vision–language modeling are established research directions, and recent work studies their intersection. The combination still raises questions that a single favorable run cannot answer: how performance changes with client skew, whether anti-collapse components help, whether fusion rankings survive seed variation, and what additional memory and communication the multimodal branch incurs.

We study these questions through OmniMed-FL, a controlled five-class proxy benchmark. Its public radiographs and synthetic clinical-style notes are not patient-matched data, so the work evaluates the learning system rather than clinical diagnosis. This scope permits internally matched comparisons while making the boundary between a systems result and a clinical claim explicit. Here, "robust" denotes stress testing across partitions, client counts, and missing modalities, not certified clinical robustness.

### A. Contributions

The main *contributions* of this paper are as follows:

- Operational runs begin from public pretrained encoders and random task modules; only the explicitly labeled warm-start ablation uses a pooled checkpoint.
- We evaluate heterogeneity, client count, anti-collapse components, initialization, and eight fusion rules under one multi-round FL protocol. Separate matched adaptations cover FedMME's one-shot and P-FIN's missing-text settings.
- We report realized client skew, GPU-synchronized training/aggregation time, peak allocated GPU memory, and formula-derived FP32 model-state volume, and evaluate retrieval over the full held-out split rather than five queries.

## II. Related Work

Centralized radiograph classifiers [1], [2] and Vision Transformers [7], [8] provide established image-only precedents. The clinical FL studies cited here are also image-only: Sheller *et al.* evaluate federated averaging, Dou *et al.* conduct multinational external validation, and Kaissis *et al.* combine differential privacy, secure aggregation, and encrypted inference [4]–[6]. FedProx and SCAFFOLD address heterogeneous-client optimization through proximal regularization and control variates, respectively [9], [10].

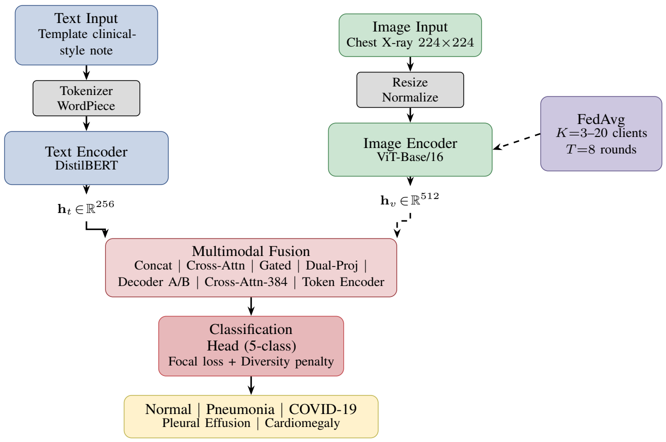


Fig. 1. OmniMed-FL architecture. DistilBERT and ViT-Base/16 encode the two proxy modalities; lightweight multimodal fusion combines them; FedAvg coordinates simulated clients without exchanging their raw examples.

BiomedCLIP and MedCLIP learn medical image–text representations, while LLaVA-Med instruction-tunes a biomedical vision–language assistant; all three are centralized [11]–[13]. Recent multimodal FL includes FedMME's one-shot hard-voting ensemble and P-FIN's uncertainty-aware feature imputation for missing modalities [14], [15]. Because their published datasets, backbones, tasks, and budgets differ, we report clearly labeled adaptations using our common corpus and local-epoch budget; we neither compare cross-paper scores nor claim exact reproduction.

## III. System Design

### A. Architecture

Figure 1 shows the OmniMed-FL pipeline. Clinical-style notes are tokenized with WordPiece to 128 tokens; chest images are resized to 224×224 and ImageNet-normalized. The `[CLS]` representation from DistilBERT [16] is projected to $\mathbf{h}_t \in \mathbb{R}^{256}$, while the mean-pooled Vision Transformer (ViT) token sequence is projected to $\mathbf{h}_v \in \mathbb{R}^{512}$.

The two representations are combined by eight lightweight operators: projected concatenation, residual cross-attention, gated residual fusion, dual-projection concatenation, one-layer decoder A, a distinct two-query/two-layer decoder B, non-residual 384-dimensional cross-attention, and a two-token Transformer encoder. Historical code keys resemble foundation-model names, but these are compact fusion rules rather than implementations of those models. Each feeds the same two-layer, five-class multilayer perceptron (MLP), and all eight are compared inside the operational FL loop with the same data, partition, optimizer, and local budget.

### B. Federated and Local Objectives

Let $K$ clients hold disjoint shards $\mathcal{D}_k$ of the training split, with $n_k = |\mathcal{D}_k|$ and $n = \sum_k n_k$, and let $\theta$ collect the trainable parameters on the fusion path: both encoders, the projection layers, the fusion operator and the classification head. The federated objective is the sample-weighted empirical risk

$$\begin{aligned} \min_{\theta}\ F(\theta) &= \sum_{k=1}^{K} \frac{n_k}{n} F_k(\theta), \\ F_k(\theta) &= \frac{1}{n_k} \sum_{(x,z,y)\in\mathcal{D}_k} \ell\big(f_\theta(x,z), y\big). \end{aligned} \tag{1}$$

where $x$ is a radiograph, $z$ is its class-paired note, $y$ is the class label, and $f_\theta$ is the fused five-class classifier. Shards are never exchanged; the server receives only the trainable tensors enumerated above.

The local loss combines a focal term with a batch-level diversity penalty,

$$\ell = (1-p_y)^{\gamma}\, \mathrm{CE}_\epsilon\big(f_\theta(x,z), y\big) \;+\; \lambda\Big(1 - \tfrac{H(\bar{p})}{\log C}\Big), \tag{2}$$

where $p_y$ is the predicted probability of the true class, $\mathrm{CE}_\epsilon$ is label-smoothed cross-entropy, $H$ is entropy, $\gamma$=2, $\epsilon$=0.05, $C$=5, and $\bar{p}$ is the mean predicted distribution over the mini-batch. The second term grows when a client's predictions concentrate on few classes, a behavior that can occur under severe Dirichlet skew. Setting $\lambda$=0 removes the diversity penalty; together, the penalty and a class-balanced mini-batch sampler form the anti-collapse stack ablated in Section IV.

At each round, the server broadcasts $\theta^{(t)}$; each eligible client performs three local AdamW epochs using class-balanced mini-batches and (2), clips the gradient norm at one, and returns its trainable tensors. The server applies the sample-weighted update in (1). Operational runs begin with public DistilBERT and ViT-Base/16 encoder weights plus random projection, fusion, and classification heads; no pooled-client model initializes them. Both encoders are fine-tuned, whereas unused unimodal heads and the inactive CNN fallback are excluded from optimization and communication.

### C. Controlled Proxy Corpus

**Image data.** We form a balanced 3,000-image proxy corpus (600 examples per class). The public radiographs come from the Kermany pneumonia collection [17] and the NIH ChestX-ray dataset [18]. Loader logs verify 2,400 public images covering Normal, Pneumonia, Pleural Effusion and Cardiomegaly. The originally specified COVID-19 source was unavailable at execution time, so the 600 COVID-19 images are synthetic X-ray-style surrogates with bilateral opacity patterns. Images are resized to 224×224, normalized with ImageNet statistics, shuffled deterministically and split 80/20 into 2,400 training and 600 validation examples.

**Clinical text data.** No patient electronic health record (EHR) data are used. Each image label is paired at the class level with a synthetic clinical-style note generated from observation, symptom, context and indicator slots. Twenty-five percent of notes draw all slots from the target class, 35% mix target and non-target slots, and 40% are heavily mixed while retaining a probabilistic target-class indicator. This controlled construction reduces direct label wording, but the image and note are not from the same patient and must not be described as a real image–report pair.

**What this corpus does and does not support.** Every experimental arm sees the same controlled corpus, which supports internally controlled, descriptive comparisons between federated algorithms, fusion rules and ablations. The absolute scores are not diagnostic-performance estimates: class-conditioned templates can create shortcuts, label-level pairing removes real patient discordance and missingness, and the synthetic COVID-19 surrogates may not resemble clinical COVID-19 radiographs. The split is also not patient-grouped across source institutions. We therefore make no claim about clinical utility, deployment readiness, or generalization to real paired reports.

### *D. Experimental Setup*

**Hyperparameters.** Each client uses AdamW with learning rate $10^{-4}$ and weight decay 0.01 for three local epochs per round. The reviewer suite uses eight rounds, batch size 16, FP32 tensors without automatic mixed precision (AMP), and a diversity-penalty weight of 1.0. The reference setting is five clients with Dirichlet $\alpha = 1.0$; Section IV varies $\alpha$ and the number of clients explicitly. Seeds 0 and 1 jointly control task-head initialization and the realized client partition. Experiments run in PyTorch on one NVIDIA H100 NVL device in a server shared with unrelated workloads. The GPU-synchronized timer covers sequential local training and aggregation within each round; validation, setup, I/O, and networking are excluded, while server contention remains uncontrolled. Deterministic CUDA algorithms are not enforced, so seeds control software pseudorandom streams without guaranteeing bitwise-identical reruns. Simulated clients train sequentially on one GPU, so time is an execution-cost record rather than distributed throughput. Table I records the fixed protocol.

**Partitioning.** For each class, a Dirichlet draw allocates training indices across clients. Smaller $\alpha$ creates more concentrated client label distributions; $\alpha \to \infty$ approaches independent and identically distributed (IID) allocation [19]. Because nominal $\alpha$ does not fully describe a finite realized split, we also report the mean client share occupied by its dominant class. The same split is held fixed across rounds for a given seed. Shards with fewer than four training examples are not locally updated; scalability results therefore report realized active and nominal $K$.

**Matched recent-method controls.** Our matched FedMME-style arm uses 24 local epochs rather than FedMME's native 100, uploads each client model once, and applies its equal-weight hard voting; our implementation breaks ties by mean softmax [14]. This preserves the common corpus, shards, encoders, optimizer, and matched local compute, but substitutes our five-class inputs for FedMME's generated-report pipeline. The P-FIN-style stress test fixes three multimodal and two image-only clients, projects both encoders to 256-dimensional normalized features, and compares zero filling, deterministic FIN, Gaussian $\beta$-negative-log-likelihood ($\beta$-NLL) imputation ($\beta$=0.5), and uncertainty-weighted aggregation with Fed-UQ-Avg balance weight 0.6 and temperature 0.2 [15]. It uses the common concatenation classifier rather than P-FIN's native fusion stack. FedProx uses $\mu$=0.01. The FedMME- and P-FIN-style controls are matched adaptations, not exact reproductions; their native scopes and communication schedules differ.

TABLE I
COMMON SIMULATION PARAMETERS.

| Parameter | Value |
|---|---|
| Local Epochs ($E$) | 3 |
| Federated Rounds ($T$) | 8 |
| Batch Size | 16 |
| AdamW Learning Rate | $1 \times 10^{-4}$ |
| Weight Decay | 0.01 |
| Diversity weight ($\lambda$) | 1.0 |
| Systems-grid $K$ (reference) | 3, 5, 10, 20 (5) |
| Systems-grid $\alpha$ (reference) | 0.1, 1, 5 (1) |
| $K$=5 additional $\alpha$ | 0.3, 0.5 |
| Seeds | 0, 1 |
| Initialization | Public encoders + random task modules |
| Precision | FP32 tensors (no mixed precision) |

**Rationale for Table I.** Within each ablation or comparison family, only the named factor changes. Three local epochs and eight rounds give a tractable 24-local-epoch budget, while batch size 16 holds both encoders and one local model copy in GPU memory. The Cartesian systems grid crosses $K$=$\{3, 5, 10, 20\}$ with $\alpha$=$\{0.1, 1, 5\}$; the $K$=5 heterogeneity sweep additionally includes $\alpha$=0.3 and 0.5. Thus the study spans severe to near-IID label skew, with $K$=5, $\alpha$=1 as its reference. With only two seeds, the results support descriptive claims but not claims of statistical significance. The $10^{-4}$ rate and $\lambda$=1 are fixed implementation defaults, not per-model optima.

## IV. RESULTS

### *A. Matched Performance and Ablations*

At $\alpha$=0.1, local-only training averages a macro-F1 score of 0.310 across five clients in one partition. Across two matched seeds, FedAvg obtains $0.652 \pm 0.057$ and FedProx obtains $0.685 \pm 0.064$; their 0.033 gap is smaller than both sample standard deviations (Fig. 2(a)). The SCAFFOLD-style AdamW adaptation obtains $0.192 \pm 0.129$; its oscillation in panel (b) indicates instability in our AdamW control-variate adaptation, not classical SGD-based SCAFFOLD. The FedMME-style arm trains five client models independently for 24 epochs and combines predictions by equal-weight hard voting with mean-softmax tie-breaking; it preserves the one-shot idea but is not an exact reproduction.

Under severe skew, the full class-balanced-sampler/diversity-penalty stack reaches $0.716 \pm 0.014$ F1; removing the sampler, diversity penalty, or both gives $0.585 \pm 0.148$, $0.701 \pm 0.011$, and $0.657 \pm 0.057$, respectively (panel (c)). Diversity diamonds distinguish transient loss of predicted-class diversity from lower F1; with only two seeds, no consistent directional component effect is evident at $\alpha$=1. Independent baseline, heterogeneity-sweep, and ablation runs nominally repeat the severe-skew FedAvg/concatenation setting but yield $0.652 \pm 0.057$, $0.674 \pm 0.185$, and $0.716 \pm 0.014$; these runs are reported separately rather than pooled or selected. Panel (d) trains all eight fusion rules in the operational FL loop.

Fig. 2. Matched results (two-seed mean±sample standard deviation unless noted). (a) Severe-skew baselines: local-only is the five-client mean from one partition; FedMME-style is a one-shot hard-voting ensemble after 24 independent local epochs, whereas iterative arms use 8 rounds of 3 epochs. (b) Iterative convergence. (c) Anti-collapse F1 with diamonds for minimum predicted-class diversity. (d) Eight operational fusion rules. (e) Federated initialization (round-1 diamonds) and a separated, non-federated 12-epoch pooled abort control. (f) P-FIN-style adaptation with three multimodal and two image-only clients; all held-out text is forced missing for the primary metric, while full-text F1 is secondary.

The all-random arm retains the DistilBERT and ViT architectures but randomly initializes all encoder and task-module weights. All three initialization arms then use the same $8{\times}3$ FL schedule; pooled-start alone reuses one seed-0 checkpoint selected by a 12-epoch centralized pooled run for both FL seeds, so its extra pretraining is outside that budget. It starts at 0.843 rather than 0.704 round-1 F1 but ends at 0.818 rather than 0.839; only this labeled ablation uses the pooled checkpoint. The separate abort-off/on controls ($0.845{\pm}0.013$ and $0.884{\pm}0.003$) are non-federated runs using pooled data without class-balanced sampling and with the diversity penalty disabled. Every run completes 12 epochs with minimum diversity 1.0, so the learning-rate boost/abort branch never fires; the difference is interpreted as rerun variation rather than an early-abort effect. Panel (f) is a matched P-FIN-style adaptation: 256-dimensional $L_2$-normalized features and concatenation plus an MLP replace the source bidirectional-attention stack; forced-missing text is primary and full-text performance secondary.

### *B. Scalability, Communication, and Resource Cost*

Figure 3(a) reports the Cartesian product of $K{=}\{3,5,10,20\}$ and $\alpha{=}\{0.1,1,5\}$. At $K{=}5$, F1 falls from $0.896\pm0.004$ at lower skew ($\alpha{=}5$) to $0.674{\pm}0.185$ under severe skew; auxiliary $\alpha{=}0.3$ and 0.5 checks yield $0.769\pm0.043$ and $0.824\pm0.113$. This spread and the realized allocation in panel (f) show why nominal $\alpha$ and one seed are insufficient. Cells report $A/K$ (or the two-seed $A$ range); shards with fewer than four samples skip local optimization.

Panels (b)–(c) report the runtime and peak allocated memory of the GPU-synchronized sequential local-training and aggregation block on one shared H100 NVL; validation, setup, input/output operations, networking, concurrency, and multi-node throughput are excluded. For $|\theta|$ trainable parameters, $b$ bytes/parameter, $T$ rounds, and nominal $K$, the plotted bidirectional volume is

$$V_{\mathrm{nom}} = 2KT|\theta|b, \tag{3}$$

Here $|\theta|{=}$153,935,621, $b{=}4$, and $T{=}8$, so a 615,742,484-byte model state yields a nominal bidirectional volume of

Fig. 3. Scalability/resource audit. Two-seed mean±sample standard deviation is shown for (a) Macro F1 ($A/K$ is active/nominal clients or the two-seed $A$ range), (b) the GPU-synchronized training/aggregation section per round, and (c) peak allocated memory within it. (d) Formula FP32 volume counts one download and upload for every nominal client in eight rounds, including locally skipped shards; it is not measured traffic. (e) One-run branch cost at $\alpha{=}1, K{=}5$ (timed-round sum and one-way model-state size). (f) Severe-skew seed-0 counts, colored by within-client class share.

27.5/45.9/91.8/183.5 GiB for $K{=}3/5/10/20$ (1 GiB= $2^{30}$ bytes). This nominal count remains when $A{<}K$; replacing $K$ by $A$ is an alternative estimate, not a measurement. Serialization, headers, retransmission, secure aggregation, compression, and algorithm-specific state are excluded.

In the one-run branch audit (panel (e)), text/image/multimodal F1 is 0.893/0.740/0.782; one-way FP32 model-state sizes are 0.248/0.324/0.573 GiB, sums of timed round durations are 18.5/33.6/45.0 min, and peak allocated memory is 1.69/3.18/4.77 GiB. The text branch outperforming multimodal concatenation shows that this proxy run does not support blanket multimodal superiority; class-conditioned template shortcuts are one plausible cause.

### *C. Retrieval-Stage Evaluation for Retrieval-Augmented Generation (RAG)*

Exact inner-product FAISS search [20] over TF–IDF note vectors yields condition-macro averages of 0.522 for top-1 same-label retrieval accuracy, 0.462 for same-label precision@5, and 0.707 for mean top-1 cosine similarity; the five condition rows cover all 600 held-out queries (Fig. 4). This synthetic-note retrieval check is neither a fine-tuned DistilBERT result nor a generated-answer evaluation, and it does not establish clinical retrieval quality.

Fig. 4. Exact retrieval using term frequency–inverse document frequency (TF–IDF) vectors and FAISS over all 600 held-out queries. Bars show top-1 same-label retrieval accuracy and same-label precision@5; diamonds show mean top-1 cosine similarity with standard-deviation whiskers. This evaluates retrieval, not generation.

## V. Discussion and Limitations

The experiments support systems-level conclusions only. Strong partition sensitivity persists despite anti-collapse measures: at $\alpha{=}0.1$, even identically configured FedAvg reruns vary substantially, and the audited runner skips local optimization for shards with fewer than four samples. The client-count study is a sequential simulation; it does not model concurrent workers, network latency, stragglers, intermittent dropout, or heterogeneous accelerators. Robust aggregation against poisoned or Byzantine updates is also untested. Accordingly, "robust" in the title denotes the reported stress tests, not a formal guarantee.

Communication is a second boundary. Equation (3) exposes model-state scaling, while Fig. 3(e) shows the larger multimodal model state and memory footprint. A deployment study must additionally measure end-to-end traffic and latency under compression, partial participation, failures, and secure aggregation. This simulation implements neither secure aggregation nor the differentially private training and privacy accounting of Abadi *et al.* [21]. It also does not evaluate attacks on model updates; retaining raw examples locally is therefore not a formal privacy guarantee.

Clinical interpretation has additional limitations. All 3,000 notes and all 600 COVID-19 images are synthetic; the other 2,400 images are public radiographs. Image–text pairing is by class rather than patient; class-conditioned templates can encode label shortcuts; and label-level pairing removes the discordance, missingness, temporal context, and site effects of real records. The split is grouped by neither patient nor source institution, and there is no external or prospective validation. Thus neither absolute F1 nor retrieval scores estimate diagnostic safety, generalization, clinical utility, or deployment readiness.

Because only two seeds were run and deterministic CUDA algorithms were not enforced, the reported means and sample standard deviations are descriptive and do not support claims of statistical significance. The matched FedMME- and P-FIN-style arms retain selected method components under a common corpus and 24-local-epoch budget, but use different architectures and schedules and substitute our five-class inputs and fusion stack; they are not exact reproductions of the source papers. Larger multi-site datasets, more seeds, native-method replications, calibrated uncertainty, privacy/security tests, and prospective evaluation are necessary next steps.

## VI. Conclusion

OmniMed-FL is a controlled systems study of multimodal FL rather than a clinical validation. Its reported performance and resource measurements are conditional on partition realization and protocol choices; the fusion, initialization, and missing-modality comparisons are descriptive stress tests. Real patient-paired multi-institutional studies and deployment safeguards remain necessary before clinical use.


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## References


[1] P. Rajpurkar, J. Irvin, K. Zhu, et al., "CheXNet: Radiologist-level pneumonia detection on chest X-rays with deep learning," *arXiv:1711.05225*, 2017.

[2] J. Irvin, P. Rajpurkar, M. Ko, et al., "CheXpert: A large chest radiograph dataset with uncertainty labels and expert comparison," in *Proc. AAAI Conf. Artificial Intelligence*, vol. 33, no. 1, pp. 590–597, 2019.

[3] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. Agüera y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. 20th Int. Conf. Artificial Intelligence and Statistics (AISTATS)*, PMLR, vol. 54, pp. 1273–1282, 2017.

[4] M. J. Sheller, B. Edwards, G. A. Reina, et al., "Federated learning in medicine: facilitating multi-institutional collaborations without sharing patient data," *Scientific Reports*, vol. 10, art. 12598, 2020.

[5] Q. Dou, T. Y. So, M. Jiang, et al., "Federated deep learning for detecting COVID-19 lung abnormalities in CT: a privacy-preserving multinational validation study," *npj Digital Medicine*, vol. 4, art. 60, 2021.

[6] G. Kaissis, A. Ziller, J. Passerat-Palmbach, et al., "End-to-end privacy preserving deep learning on multi-institutional medical imaging," *Nature Machine Intelligence*, vol. 3, pp. 473–484, 2021.

[7] A. Dosovitskiy, L. Beyer, A. Kolesnikov, et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2021.

[8] Z. Liu, Y. Lin, Y. Cao, et al., "Swin Transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2021, pp. 10012–10022.

[9] T. Li, A. K. Sahu, M. Zaheer, et al., "Federated optimization in heterogeneous networks," in *Proc. Mach. Learn. Syst. (MLSys)*, vol. 2, pp. 429–450, 2020.

[10] S. P. Karimireddy, S. Kale, M. Mohri, et al., "SCAFFOLD: Stochastic controlled averaging for federated learning," in *Proc. 37th Int. Conf. Machine Learning (ICML)*, PMLR, vol. 119, pp. 5132–5143, 2020.

[11] S. Zhang, Y. Xu, N. Usuyama, et al., "BiomedCLIP: A multimodal biomedical foundation model pretrained from fifteen million scientific image-text pairs," *arXiv:2303.00915*, 2023.

[12] C. Li, C. Wong, S. Zhang, et al., "LLaVA-Med: Training a large language-and-vision assistant for biomedicine in one day," in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, vol. 36, pp. 28541–28564, 2023.

[13] Z. Wang, Z. Wu, D. Agarwal, and J. Sun, "MedCLIP: Contrastive learning from unpaired medical images and text," in *Proc. Conf. Empirical Methods in Natural Language Processing (EMNLP)*, 2022, pp. 3876–3887.

[14] N. Wang, Y. Deng, S. Fan, J. Yin, and S.-K. Ng, "Multi-modal one-shot federated ensemble learning for medical data with vision large language model," *arXiv:2501.03292*, 2025.

[15] N. F. Shahid, M. Ahmed, M. A. Haider, et al., "Probabilistic feature imputation and uncertainty-aware multimodal federated aggregation," in *Proc. Medical Imaging with Deep Learning (MIDL)*, PMLR, vol. 315, pp. 3593–3607, 2026.

[16] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, "DistilBERT, a distilled version of BERT: Smaller, faster, cheaper and lighter," *arXiv:1910.01108*, 2019.

[17] D. S. Kermany, M. Goldbaum, W. Cai, et al., "Identifying medical diagnoses and treatable diseases by image-based deep learning," *Cell*, vol. 172, no. 5, pp. 1122–1131.e9, 2018.

[18] X. Wang, Y. Peng, L. Lu, et al., "ChestX-ray8: Hospital-scale chest X-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 2097–2106.

[19] T.-M. H. Hsu, H. Qi, and M. Brown, "Measuring the effects of non-identical data distribution for federated visual classification," *arXiv:1909.06335*, 2019.

[20] J. Johnson, M. Douze, and H. Jégou, "Billion-scale similarity search with GPUs," *IEEE Trans. Big Data*, vol. 7, no. 3, pp. 535–547, 2021.

[21] M. Abadi, A. Chu, I. Goodfellow, et al., "Deep learning with differential privacy," in *Proc. ACM SIGSAC Conf. Computer and Communications Security (CCS)*, 2016, pp. 308–318.